Power Tree

XBandDigitalModule_20x20 - Power Tree Document

Document Information

Field	Value
Document ID	XBDM-DOC-002
Version	1.0
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Classification	ITAR Restricted
Status	Preliminary Des

1. Power Architecture Overview

1.1 Input Power Specification

Parameter	Min	Typ	Max	Unit	Notes
Input Voltage	4.0	4.5	5.5	V	Nominal 4.5V
Input Current	-	16.2	25	A	At 96.5W max
Input Power	-	72.7	96.5	W	Including losse
Ripple	-	-	100	mVpp	Before regulati
Transient	3.5	-	6.0	V	MIL-STD-704
Dropout	-	-	0.5	V	Min input for r

1.2 Power Rail Requirements

Rail	Voltage	Tolerance	Current	Noise	Sequencing
VCCINT	1.0V	±3%	44A	<10 mVpp	First
VCCAUX	1.8V	±3%	6A	<20 mVpp	Second
VCCO	3.3V	±3%	3A	<30 mVpp	Third
AVDD	1.0V	±2%	1.5A	<1 mVpp	Fourth
DVDD	1.8V	±3%	1.5A	<5 mVpp	Fifth
VCLK	2.5V	±3%	2A	<2 mVpp	Fourth

2. Power Tree Diagram

```
                          +--------------------------------------------------------------+
                          |                    POWER INPUT SECTION                        |
                          |                                                              |
                          |  +-------------+   +-------------+   +-------------+       |
 4.5V Input -----------?|  | Reverse     |--?| Input       |--?| EMI         |-------+|
 (4.0-5.5V)               |  | Polarity    |   | Filter      |   | Filter      |       ||
                          |  | Protection  |   | (Pi filter) |   | (Common     |       ||
                          |  +-------------+   +-------------+   |  mode)      |       ||
                          |                                     +-------------+       ||
                          +--------------------------------------------------------------+
                                                                                    |
                                                                                    ?
                          +--------------------------------------------------------------+
                          |                    POWER DISTRIBUTION                        |
                          |                                                              |
                          |  +----------------------------------------------------+   |
                          |  |               4.5V BUS (after protection)           |   |
                          |  +----------------------------------------------------+   |
                          |         |         |         |         |         |           |
                          |         ?         ?         ?         ?         ?           |
```

```
|    +--------++--------++--------++--------++--------+     |
|    | BUCK 1 || BUCK 2 || BUCK 3 || LDO 1  || LDO 2  |     |
|    | TPS7H  || ISL70  || ISL70  || TPS7H  || TPS7H  |     |
|    | 5002   || 003A   || 003A   || 1111   || 1111   |     |
|    +---+----++---+----++---+----++---+----++---+----+     |
|        |         |         |         |         |          |
|        ?         ?         ?         ?         ?          |
|    +--------++--------++--------++--------++--------+     |
|    | 1.0V   || 1.8V   || 3.3V   || 1.0V   || 1.8V   |     |
|    | Core   || Aux    || I/O    || Analog || Digital|     |
|    | 44A    || 6A     || 3A     || 1.5A   || 1.5A   |     |
|    +--------++--------++--------++--------++--------+     |
|                                                             |
|  +-------------------------------------------------------+   |
|  |              2.5V LDO (from 3.3V)                     |   |
|  |              TPS7H1121-SP                              |   |
|  |              2A                                        |   |
|  +-------------------------------------------------------+   |
|                                                             |
|  +-------------------------------------------------------+   |
|  |              POWER SEQUENCER                          |   |
|  |              TPS7H3014-SP (4-channel)                 |   |
|  |              Controls: VCCINT->VCCAUX->VCCO->AVDD->DVDD|   |
|  +-------------------------------------------------------+   |
|                                                             |
|  +-------------------------------------------------------+   |
|  |              POWER MONITORING                         |   |
|  |              INA214-SP (current sense)                |   |
|  |              TPS7H3014-SP (voltage monitor)           |   |
|  +-------------------------------------------------------+   |
```

3. Detailed Rail Analysis

3.1 VCCINT (1.0V, 44A) - FPGA Core

Component: TPS7H5002-SP (Buck Controller) + External MOSFETs

Parameter	Value	Notes
Input Voltage	4.5V (4.0-5.5V)	Direct from bus
Output Voltage	1.0V ±3%	0.97V - 1.03V
Output Current	44A max	35A typical
Switching Frequ	500 kHz	Configurable
Efficiency	92%	At 35A, 4.5V in
Output Ripple	<10 mVpp	With ceramic ca
Load Transient	<50 mV	0-100% load ste
Thermal	2.8W loss	At 92% efficien

External Components:

Protection:

3.2 VCCAUX (1.8V, 6A) - FPGA Auxiliary

Component: ISL70003ASEH (Integrated Buck)

Parameter	Value	Notes
Input Voltage	4.5V (4.0-5.5V)	Direct from bus
Output Voltage	1.8V ±3%	1.75V - 1.85V
Output Current	6A max	4A typical
Switching Frequ	500 kHz	Fixed
Efficiency	94%	At 4A, 4.5V inp
Output Ripple	<15 mVpp	With ceramic ca
Load Transient	<30 mV	0-100% load ste
Thermal	0.45W loss	At 94% efficien

Internal MOSFETs: 31 m? (high-side), 21 m? (low-side)

External Components:

3.3 VCCO (3.3V, 3A) - I/O and Clock

Component: ISL70003ASEH (Integrated Buck)

Parameter	Value	Notes
Input Voltage	4.5V (4.0-5.5V)	Direct from bus
Output Voltage	3.3V ±3%	3.20V - 3.40V
Output Current	3A max	2A typical
Switching Frequ	500 kHz	Fixed
Efficiency	93%	At 2A, 4.5V inp
Output Ripple	<20 mVpp	With ceramic ca
Load Transient	<40 mV	0-100% load ste
Thermal	0.47W loss	At 93% efficien

3.4 AVDD (1.0V, 1.5A) - ADC Analog

Component: TPS7H1111-SP (Ultra-Low Noise LDO)

Parameter	Value	Notes
Input Voltage	3.3V	From VCCO via f
Output Voltage	1.0V ±2%	0.98V - 1.02V
Output Current	1.5A max	1.2A typical
Dropout Voltage	200 mV	At 1.5A
Output Noise	1.71 μVRMS	10 Hz - 100 kHz
PSRR	71 dB	@ 100 kHz
Thermal	0.45W loss	At 3.3V input,

External Components:

Critical: This rail powers ADC analog circuitry. Ultra-low noise is essential for maintaining SNR.

3.5 DVDD (1.8V, 1.5A) - ADC Digital

Component: TPS7H1111-SP (Ultra-Low Noise LDO)

Parameter	Value	Notes
Input Voltage	3.3V	From VCCO via f
Output Voltage	1.8V ±3%	1.75V - 1.85V
Output Current	1.5A max	1.0A typical
Dropout Voltage	200 mV	At 1.5A
Output Noise	1.71 μVRMS	10 Hz - 100 kHz
PSRR	71 dB	@ 100 kHz
Thermal	0.75W loss	At 3.3V input,

3.6 VCLK (2.5V, 2A) - Clock and Balun

Component: TPS7H1121-SP (General Purpose LDO)

Parameter	Value	Notes
Input Voltage	3.3V	From VCCO
Output Voltage	2.5V ±3%	2.43V - 2.58V
Output Current	2A max	1.5A typical
Dropout Voltage	300 mV	At 2A
Output Noise	4 μVRMS	10 Hz - 100 kHz
PSRR	45 dB	@ 100 kHz
Thermal	1.2W loss	At 3.3V input,

4. Power Sequencing

4.1 Sequencing Requirements

Order	Rail	Delay	Reason
1	VCCINT (1.0V)	0 ms	FPGA core must
2	VCCAUX (1.8V)	10 ms	Auxiliary befor
3	VCCO (3.3V)	20 ms	I/O banks
4	AVDD (1.0V)	30 ms	ADC analog (aft
5	DVDD (1.8V)	40 ms	ADC digital (af

4.2 Sequencer Implementation

Component: TPS7H3014-SP (4-Channel Sequencer)

```
Channel 1: VCCINT (1.0V) --? Delay 10ms --?
Channel 2: VCCAUX (1.8V) --? Delay 10ms --?
Channel 3: VCCO (3.3V) --? Delay 10ms --?
Channel 4: AVDD (1.0V) --? (external RC for DVDD delay)
```

PGOOD Chain:

4.3 Shutdown Sequence

Reverse order: DVDD -> AVDD -> VCCO -> VCCAUX -> VCCINT

Emergency Shutdown: If any rail faults, all rails shutdown simultaneously via FAULT pin.

5. Protection Circuits

5.1 Input Protection

Function	Component	Specification
Reverse Polarit	P-MOSFET (IRF93	-40V, 6.5A, RDS
OVP	TVS (SMDJ6.0A)	6.0V breakdown,
Inrush Current	NTC Thermistor	5? cold, 0.3? h
EMI Filter	Pi-filter	2x 10µH + 2x 10
Fuse	Resettable (Bou	5A hold, 10A tr

5.2 Per-Rail Protection

Rail	OVP	OCP	UVLO	Thermal
VCCINT	1.15V (115%)	48A (109%)	0.85V (85%)	150 degC
VCCAUX	1.95V (108%)	7A (117%)	1.5V (83%)	150 degC
VCCO	3.5V (106%)	3.5A (117%)	2.8V (85%)	150 degC
AVDD	1.05V (105%)	2A (133%)	0.8V (80%)	125 degC
DVDD	1.9V (106%)	2A (133%)	1.5V (83%)	125 degC
VCLK	2.65V (106%)	2.5A (125%)	2.0V (80%)	125 degC

5.3 Fault Handling

Fault Sources:

1. 2.

Fault Response:

6. Power Monitoring

6.1 Voltage Monitoring

Rail	Monitor	Threshold	Method
VCCINT	TPS7H3014-SP	±5%	Internal compar
VCCAUX	TPS7H3014-SP	±5%	Internal compar
VCCO	TPS7H3014-SP	±5%	Internal compar
AVDD	INA214-SP	±3%	External sense
DVDD	INA214-SP	±5%	External sense

6.2 Current Monitoring

Rail	Monitor	Range	Accuracy
Input (4.5V)	INA214-SP	0-25A	±1%
VCCINT	INA214-SP	0-50A	±1%
VCCAUX	INA214-SP	0-10A	±2%
Total Power	Calculated	0-100W	±3%

6.3 Temperature Monitoring

Location	Sensor	Range	Accuracy
FPGA die	Internal DTS	-40 to +125 deg	±3 degC
ADC die	Internal	-40 to +125 deg	±5 degC
PCB hot spot	TMP102-SP	-40 to +125 deg	±1 degC
Ambient	TMP102-SP	-40 to +85 degC	±1 degC

7. Efficiency Analysis

7.1 Rail Efficiency

Rail	Topology	Efficiency @ Ty	Efficiency @ Ma
VCCINT (1.0V)	Buck (TPS7H5002	92%	90%
VCCAUX (1.8V)	Buck (ISL70003)	94%	92%
VCCO (3.3V)	Buck (ISL70003)	93%	91%
AVDD (1.0V)	LDO (TPS7H1111)	30%	30%
DVDD (1.8V)	LDO (TPS7H1111)	55%	55%
VCLK (2.5V)	LDO (TPS7H1121)	76%	76%

7.2 Total System Efficiency

```
Buck Rails:
  VCCINT: 60W x 92% = 55.2W output, 6.8W loss
  VCCAUX: 10.8W x 94% = 10.1W output, 0.7W loss
  VCCO:   9.9W x 93% = 9.2W output, 0.7W loss

LDO Rails:
  AVDD: 1.5W x 30% = 1.5W output, 3.5W loss
  DVDD: 2.7W x 55% = 2.7W output, 2.2W loss
  VCLK: 5W x 76% = 5W output, 1.6W loss

Total Output Power: 83.7W
Total Input Power: 96.5W
Overall Efficiency: 86.7%
```

8. Thermal Considerations

8.1 Power Dissipation by Component

Component	Power Loss	Package	?JC	?JA	TJ Max
TPS7H5002-SP	0.3W	CFP-22	15 degC/W	50 degC/W	125 degC
External MOSFET	2.8W	D2PAK	1.5 degC/W	40 degC/W	150 degC
ISL70003ASEH (x	0.9W	CQFP-64	5 degC/W	25 degC/W	150 degC
TPS7H1111-SP (x	1.2W	HTSSOP-28	3 degC/W	30 degC/W	125 degC
TPS7H1121-SP	1.2W	CFP-22	4 degC/W	35 degC/W	125 degC
TPS7H3014-SP	0.1W	CFP-22	10 degC/W	45 degC/W	125 degC
Total	**6.5W**	-	-	-	-

8.2 Thermal Management

- **PCB Copper**: 2 oz copper on power layers

9. EMC Considerations

9.1 Conducted Emissions

Frequency	Limit	Mitigation
150 kHz - 30 MH	CISPR 11 Class	Pi-filter at in
30 MHz - 1 GHz	CISPR 11 Class	Common-mode cho

9.2 Radiated Emissions

Frequency	Limit	Mitigation
30 MHz - 1 GHz	CISPR 11 Class	Shielded enclos
1 GHz - 6 GHz	CISPR 11 Class	PCB layout, gro

9.3 PCB Layout Guidelines

- **Ground Plane**: Continuous ground plane on layer 2

10. BOM Summary

10.1 Power Section BOM

Reference	Part Number	Description	Qty	Unit Cost	Total
U1	TPS7H5002-SP	Buck Controller	1	$45.00	$45.00
U2	ISL70003ASEH	Integrated Buck	2	$35.00	$70.00
U3	TPS7H1111-SP	Ultra-Low Noise	2	$25.00	$50.00
U4	TPS7H1121-SP	General LDO	1	$20.00	$20.00
U5	TPS7H3014-SP	Sequencer	1	$30.00	$30.00
U6	INA214-SP	Current Sense A	3	$15.00	$45.00
Q1-Q4	CSD19538Q3A	MOSFET 100V 3.8	4	$8.00	$32.00
L1	XAL6060-402	Inductor 4µH 60	1	$12.00	$12.00
L2-L3	XAL5030-222	Inductor 2.2µH	2	$6.00	$12.00
C1-C36	C3225X5R	Capacitors 10µF	36	$0.50	$18.00
**Total Power S	-	-	**53**	-	**$334.00**

10.2 Cost Estimation

Category	Cost	Notes
Power Section	$334	As above
ADC Section	$850	ADC12DJ5200-SP
FPGA Section	$180,000	XQRVC1902 + DDR
Clock Section	$450	LMX2615-SP + LM
Mechanical	$2,500	PCB + chassis +
Assembly	$5,000	SMT + test
**Total Module*	**~$189,134**	-

11. Risk Assessment

Risk ID	Description	Probability	Impact	Mitigation
P001	Buck regulator	Low	High	Proper compensa
P002	LDO thermal shu	Medium	Medium	Derating, therm
P003	Sequencing timi	Low	High	Programmable de
P004	Noise coupling	Medium	High	Separate rails,
P005	Input voltage t	Low	Medium	TVS protection,

12. Revision History

Version	Date	Author	Description
1.0	2026-09-09	Power Electroni	Initial release
